



Digital methods III: The digital mundane

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Abstract

This report considers a burgeoning strand of scholarship that foregrounds the mundane in engagements with the digital. Research concerned with the digital mundane attends to the ordinary and often taken-for-granted digital objects, practices, productions, and sites that significantly both mediate and are mediated by everyday lives and spatialities. Methodological innovations are advancing new techniques for researching mundane digital objects that participate in the internet of things, everyday spaces of the smart home, banal landscapes of data and digital infrastructures, and quotidian quantifications/datafications of the self. The proliferation of these methods also informs feminist scholarly praxes of digital iteration.

Keywords

digital methods, everyday, internet of things, mundane, quantified self, smart home

I Introduction

A substantial emphasis of research in the burgeoning field of digital geographies is on newly emergent phenomena such as cryptocurrencies (Crandall, 2019), on-demand meal delivery platforms (Richardson, 2019), and RealTech¹ (Dal Maso et al., 2019). Much of this work foregrounds the ability of these technologies to supplant old configurations with new (e.g. the ‘automated landlord’ replacing interpersonal relations between property owners and tenants; Fields, 2019) and often skews towards highlighting the deleterious consequences of such digital disruptions (for instance, the alienation of labour across sectors of the gig economy such as ridehailing; Attoh et al., 2019). Work in this vein importantly renders visible the intensifying and accelerating influence exerted by digital technologies and technology capital across geographical scales from the hyperlocal to the planetary.

Yet against this tendency to prioritize the spectacularism and exceptionalism of the new, a significant and exciting strand of scholarship is attending instead to the ‘digital mundane’: the on-the-surface, taken-for-granted, seemingly ordinary and routine sites, objects, data productions, and networked practices of everyday life (Elwood and Leszczynski, 2018; Hjorth et al., 2018; Leszczynski and Crampton, 2016; Pink et al., 2017). These include ordinary devices with network connectivity that participate in the internet of things (IoT), the intimate spatialities of the smart home, the banal landscapes of digital infrastructures and data, quotidian quantifications and datafications of the self, and digital iteration as a political scholarly praxis.

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Attunement to the mundane implicitly recognizes broader digital-material configurations – often engaged as infra/structural – to be situated in pervasive digital mediations of the spaces and practices of daily life (Barns, 2018; Elwood and Leszczynski, 2018; Pink et al., 2017). The everyday is a key site of digital production and also one where digital data and devices are ‘lively’, i.e. where they come to matter by being invested with significance and ‘navigate[d], imagine[d.] and improvise[d] with’ in quotidian ‘routines’, extemporizations, and ‘accomplishments’ (Pink et al., 2017: 4). As such, the mundane offers perhaps the most direct entry point for substantially engaging with and beginning to understand ‘how digital technologies are implicated in and by the lives of ordinary people’ (Pink et al., 2017: 3), and of the consequences of these intensifying and accelerating implications.

Methodologically, work attending to the digital mundane is highly innovative, with scholars and practitioners in geography and corollary fields experimenting with, devising, repurposing, and advancing new methods for researching – and researching with – the internet of things (IoT), the smart home, landscapes of data and digital infrastructures, and quantified self-data productions. In the sections that follow, I report on a selection of these novel methods, which I organize around the registers of the digital mundane with which they engage: mundane materialities, mundane spaces, and mundane practices and praxes.

II Mundane materialities

Digital materialities consist of physical devices (hardware), software, and data that ultimately have a material manifestation as information stored on a server in a data centre or as sensory content accessed through digital speakers and screens. With the rise of the internet of things (IoT), a phenomenon which describes the proliferation of ‘smart’ objects that can send and

receive data and that may be controlled via wireless networks, we now live with an abundance of a diverse array of ordinary connected devices that are estimated to reach one trillion in number worldwide by 2025 (The Economist, 2019). One of the key sites where the IoT has taken hold is in the everyday spaces of what has been designated the ‘smart home’. The home has been made ‘smart’ in two important ways. The first involves the incorporation of connectivity as a native design feature of ordinary, previously non-networked household devices, including thermostats, refrigerators, and WiFi-enabled socket plugs that may be used to connect toasters, coffee makers, and other appliances. The second involves the introduction of new kinds of objects such as smart speakers and voice-activated digital home assistants – including Amazon Echo², Google Home³, and Apple Homepod⁴ – that function as ‘hubs’ for controlling these networked devices and for accessing digital content.

Several original methods have recently been developed for researching with connected objects in the home. Maalsen (2019) outlines three new methods for engaging with voice-activated digital assistants in studies of smart home assemblages: *lively object ethnographies*, *interviewing things*, and *enrolling things as co-ethnographers*. *Lively object ethnography* describes a ‘follow the things’ (Cook, 2004) approach to the Internet of Things by ‘charting the global movements and biographies’ of DPAs as commodities and commodity chains (Maalsen, 2019: 10). *Interviewing things* treats voice-interactive smart objects as interview participants by posing questions to them and analysing how they respond and the kinds of responses they return (Maalsen, 2019). For instance, interviews conducted with previous versions of Apple’s intelligent agent Siri⁵, which is coded with a woman’s voice, found that Siri responded obsequiously to derogatory, gendered language (West et al., 2019). *Enrolling things as co-ethnographers* consists of

utilizing networked devices such as smartphones as instruments through which research participants may collect their own ethnographic data in situ and in real time, including voice narratives, image and video capture, and geo-data. Personal devices such as smartphones may also be used by researchers to send participants notifications reminding or prompting them to collect data (Maalsen, 2019).

Elsewhere, through their 'Anatomy of an AI'⁶ project, Crawford and Joler (2018) developed a method for engaging mundane objects of the IoT that consists of *mapping* (visualizing) the intersecting social, environmental, and data ecologies that constitute an Amazon Echo device and its artificial intelligence (AI) agent Alexa. These include the geological processes, human labour, commodity supply chains, and environmental impacts of extracting raw materials necessary for the device's production (e.g. lithium mining); the human knowledge and labour necessary to produce the data (images, audio, video, text) used to train artificial intelligence; the data transmission infrastructures and energy consumed by the algorithms needed to actually train and optimize an AI; and the socialites and flows of data produced through mundane interactions with networked objects (e.g. Echo devices) deployed across the quotidian spaces of everyday life (the home, the office). As a method, *ecological mapping* both situates AI (as encapsulated within a mundane object, an Amazon Echo device) within planetary ecologies of human labour, data, and geological processes and raw materials, and simultaneously conceptualizes AI as an ecology in and of itself, one which enrolls and is implicated in orderings of material resources, human knowledge capacity, labour, data, and energy (Crawford and Joler, 2018).

In addition to new methods for engaging mundane connected devices and their AI agents as objects of study, a separate strand of methodological innovation engages in the *speculative design* of bespoke connected objects. Once designed, these objects are deployed as

instruments of data generation around which to centre *autoethnographies of (smart home) data* rather than of the devices themselves, which tend to be the focus in other studies such as the ones profiled immediately above. In a particularly innovative project conducted by Speed and Luger (2019), one of the researchers embedded a sensor in perhaps the most mundane object imaginable: the toilet paper roll dispenser in one of the bathrooms in their family home. This sensor logged toilet paper use (consumption and availability) and correlated it with the presence/absence of persons detected through a separate passive wireless LED sensor likewise placed in the bathroom. These data were logged to a database and published as a graph online. An autoethnographic engagement with this data *and* with the devices revealed the unanticipated 'disrupti[ons]' brought on by the datafication of the quotidian spaces of the home via both the attachment of sensors to mundane objects and the publication of the data produced through this sudden connection of previously unconnected objects (Speed and Luger, 2019: 130). The first unanticipated disruption involved the ways in which such practices of datafication render quotidian though private bodily functions visible. As Speed and Luger report (2019), the passive LED sensor provoked concerned questioning on the part of one of the researcher's children about whether or not the device could 'see' them while they were in the bathroom, concerns which the researcher was able to allay by explaining that there was no visual monitoring or recording involved. The second unanticipated effect involved disruption produced by the data themselves. An alert to a spike in the online graph showing a sudden increase in toilet paper availability on one occasion when the researcher and their family were away from home on holiday triggered an immediate concern that the home had been broken into, only for the researcher to deduce that this particular data signature corresponded to a regularly scheduled visit by the family's

housekeeper, who had changed the toilet paper roll (Speed and Luger, 2019).

III Mundane spaces

Beyond the micro-scale spatialities of the smart home foregrounded in the previous section, researchers are also developing methods for empirically engaging the banal landscapes of digital infrastructures and data across the meso- and macro-scale geographies of cities and regions. A frequent approach consists of *visiting* the often unnoticed, frequently prosaic, and sometimes intentionally obfuscated architectures of material internet infrastructures that make everyday connectivity possible. *Visiting* describes a comprehensive approach comprised of methods that include travelling to, photographing, sketching/diagramming, and otherwise documenting these sites, as well as conducting interviews with key stakeholders at important infrastructural locations. As a constellation of methods, *visiting* serves to render the spatial geographies of networked connectivity visible, orients an attunement towards often ignored and/or unnoticed landscapes of digital infrastructures, and situates the possibilities of quotidian digitally-mediated interaction within the mundane architectures and built environments of assemblages that we refer to as ‘the internet’.

Spanning Western Europe and North America, Blum (2012) visited internet exchanges, data centres, and cable landing sites, explaining the location of these infrastructural nodes in terms of the geographies of regional development, climate and energy resource availability, global finance capital, militarization, and historical core-peripheries of the telecommunications industry. Focusing specifically on deep sea cable networks across the Pacific, Starosielski (2015) visited cable landing sites across the region, drawing on similar methods to inform a historiographic geography of empire, militarization, international development, Indigenous

sovereignty, and the colonial present made material through cable infrastructures throughout the region. And, at the urban scale, Burrington (2016) likewise visited the internet infrastructures of New York City, paying particular attention to its easily-missed visual markers and using this data to produce a field guide to the city’s surface and subsurface networks, themselves often veiled by the most banal features of the urban built environment, such as paved streets and manhole covers.

Decentring material infrastructures as primary entry points for researching digital landscapes, Mahmoudi and Levenda (2019) develop *data walks* as a method for empirically engaging with the everyday landscapes of *data*, which are often dematerialized through invocations of the a-spatial metaphor of ‘the cloud’. Walking Seattle’s South Lake Union (SLU) district – a recently redeveloped neighbourhood whose rapid transformation was financed by and for technology capital – Mahmoudi and Levenda (2019) engaged with how this landscape both makes and is made by data. Focusing on mobilities and consumption, they traced how their movements through SLU are captured as data as their smartphones register against cell phone towers and WiFi masts, and are mined by numerous mobile applications that work to deduce and predict both their mode of travel (on foot, on the light rail link, on bike or on scooter) and their likely spatial trajectories (direction and path of travel). They mentally mapped how and where technology capital shows up in the neighbourhood’s architecture, such as in the large Whole Foods (purchased by Amazon in 2017) situated near the waterfront. As they walked into stores throughout the area, they noted that in some retail locations, such as a convenience store, consumer products suggested for purchase by data-driven algorithms on Amazon’s online marketplace could be delivered for pick-up to Amazon lockers that may be opened using a unique code sent to the customer (Mahmoudi and Levenda, 2019). Continuing on their walk, Mahmoudi and

Lavenda (2019) further reflected on the ways in which data are central to defining South Lake Union as a neighbourhood. One of the ways in which they identify that data make SLU is through the district's high grade of walkability, which is reported as a branded Walk Score⁷ (a data product produced by a company of the same name) that increases the neighbourhood's desirability. This walkability metric is then enrolled and circulated as a data point via a number of real estate platforms through which the neighbourhood is packaged and sold to prospective buyers (Mahmoudi and Levenda, 2019).

IV Mundane practices and praxes

In addition to the advancement and refinement of methods for engaging with the datafication of urban landscapes, researchers have also devised new approaches for engaging with quotidian datafications of the self that are part and parcel of a phenomenon that has been termed the quantified self (QS; see Lupton, 2016). Here, approaches emphasize two facets of self-datafication through engagements with QS technologies: 1) the actual processes of producing what Pink et al. (2017) refer to as 'mundane data'; and 2) practices involving subsequent interactions with one's own self-tracking data. As an example of the first of these two approaches, Pink et al. (2017) deployed a *sensory ethnography* approach to discern how self-tracking cycling commuters engage with quantification technologies – including motion sensors, bicycle computers, GPS units, and activity logging applications such as Strava – as part of their regular commuting activities. As part of this study, Pink et al. (2017) equipped their participants with hands-free video cameras that recorded their commutes, capturing how cyclists interacted with an array of logging technologies and the data they produced in situ during a recorded ride, including data points pertaining to speed, distance, and route. Pink et al. (2017) subsequently processed the video

footage, and watched the videos together with their participants, interviewing each cyclist about the particularities and significance of their interactions with both the tracking devices and the data produced by these technologies both during and immediately after the ride. Using this multi-pronged ethnographic approach, Pink et al. (2017) discerned that quotidian practices of mundane data production and their sharing via social media were central to how cyclists both experienced and understood their routine commutes as personal achievements and through which these achievements became both meaningful and material to them.

A second set of approaches focuses on the afterlives of QS data by shifting the focus to how individuals (re)purpose their own mundane data outside of the activity of tracking itself. Nafus and Sherman (2014) for instance relied on traditional social science methods of *participant observation* and *discourse analysis*, attending meet-ups of QS enthusiasts at which participants gave talks presenting visualizations of their own self-tracking data. Attending to both the data being visualized, the specific techniques of visualization, and how QS participants presented these visualizations to others at the meet-up, Nafus and Sherman (2014) found that individuals used visualization not only as a means of making sense of the datafication of their everyday lives, but also as a form of 'soft resistance' that frequently resulted in QS enthusiasts altering how they engaged with self-tracking technologies and practices (2014: 1785). These alterations in personal practices disrupted the logics of big data collection, algorithmic governance, and artificial intelligence by inverting the power dynamics of how data is produced, utilized, interpreted, and by whom.

In addition to datafication practices and mundane data featuring as subjects/objects of academic research, digital mundane practices are importantly also increasingly figuring as scholarly political *praxes* of digital *iteration*. Meghan Kelly (2019), for instance, has

described a personal feminist cartography praxis consisting of a daily iteration of digital map markers in an icon set, redesigning them by following D'Ignazio and Klein's (2016) principles for feminist data visualization. This practice consists of iterating a map marker each day, beginning with cartographic symbols starting with the letter 'A' and moving successively through the Roman alphabet one icon and one day at a time. As an example, Kelly proposed that one way in which the map symbol for American-style football – included in an icon set as a graphic of the oblong-shaped football with its characteristic stitching – could attune to feminist concerns with embodiment that would be in the form of a new or alternative icon consisting of a football helmet. A helmet map marker represents a means of bringing the body back into the icon set by more immediately symbolizing the bodies of the players as well as signalling the increase in public consciousness about the long-term effects of football-related head injuries. Likewise focusing on the digital body, Wilmott (2019) iterated a digital avatar for embodying spatial design spaces such as SketchUp⁸, a 3D modelling software, in an effort to disrupt the hegemonic heteronormativity and masculinity of digital bodies and digital spaces. Wilmott (2019) specifically iterated an originally female-presenting avatar named Stacy⁹, referring to them using gender-pronouns and giving them purple skin and fluorescent green hair befitting of their more-than-human '[d]igital-material cyborg' body. Wilmott (2019) furthermore removed their left hand, narrating this as an intentional act on Stacy's part, who 'cut off their hand in order to delete their wedding ring' – included as a design default – 'as an archaic relic of a previously patriarchal society'.⁹ Wilmott (2019) has made Stacy available for public download via Trimble's 3D Warehouse⁹, with 498 downloads to date (2 October 2019). This allows Stacy to inhabit multiple digital design environments simultaneously; in so doing, Stacy queers these spaces and allows others who enter to re-

encounter them in a feminist way (Wilmott, 2019).

V For mundane digital futures

Scholarly praxes of mundane digital iteration capture the significance of tactics as a site and modality of living and enacting a digital politics firmly rooted in the everyday. This disrupts narratives which suggest that the only two options for 'being political' in the digital present are resistance to techno-capitalism through strategies of collective action, or complicit acquiescence (see Leszczynski, 2019; Rose, 2017). Attunement to the digital mundane offers researchers a way out of this false binary. It furthermore has the potential to open up novel, as-yet-unexplored avenues for future research beyond the scope of current hegemonic tendencies to deploy the 'totalizing analytics' of capitalism and neoliberalism as entry points for engagements with the digital, which serve to do little beyond confirming 'capitalism's explanatory power' (Derickson, 2018: 577; for a recent example of a capitalist analysis of the smart home see Goulden, 2019).

The methods profiled in this progress report represent possible avenues for avoiding and/or surpassing such binaries and hegemonies. These include but are not limited to *speculative design*, *data walks*, *lively object/auto/sensory ethnography*, *interviewing things*, *mapping* AI ecologies, *visiting* digital infrastructures, and digital praxes of *iteration*. Using these methods to attend to and discern how digital materialities, spatialities, and praxes come to matter in and for the lives of everyday people is key to both recognizing people's quotidian capacities to enact meaningful social change *and* to rendering violence, injustice, and inequalities legible and tractable for intervention (Ahmed, 2017; Elwood and Leszczynski, 2018). This, Sara Ahmed (2017) reminds us, is the role and responsibility of research.

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Notes

1. RealTech is a portmanteau of 'real estate' and 'technology'. It has gained traction as a common designator for an array of digital enterprises and products in the industry.
2. On <https://www.amazon.com>, see Amazon devices > Amazon Echo & Alexa Devices.
3. See https://store.google.com/product/google_home
4. See <https://www.apple.com/homepod/>
5. See <https://www.apple.com/siri/>
6. See <https://anatomyof.ai/>
7. See <https://www.walkscore.com/>
8. See <https://www.sketchup.com/>
9. See <https://3dwarehouse.sketchup.com/model/0e970745-bb7d-45ca-aebd-5428da08be9e/Stacy>

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